

PROBLEM SET 1—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: These are my suggested solutions. It is your responsibility to check this document against your own solutions.

1. When checking whether each of the examples above satisfy the two axioms, for each example you either need to provide a proof if the example satisfies the axiom, or a counterexample if it does not.

Preference for uniformity: Note that this preference relation is represented by the utility function $U(p) = -\sum_{z \in Z} (p(z) - 1/|Z|)^2$ for every $p \in L(Z)$. Since U is continuous, it satisfies the continuity axiom. Since it can be shown that $U(p)$ is not linear, it cannot satisfy the independence axiom. However, let me just show you a counterexample that shows the violation of independence. Consider $[z_1]$ and $[z_2]$, note that $[z_1] \sim [z_2]$. However, $[z_1] = 1/2[z_1] \oplus 1/2[z_1] \prec 1/2[z_1] \oplus 1/2[z_2]$.

Preference for greatest likelihood: Note that this preference relation can be represented by the utility function $\max\{p_1, \dots, p_K\}$. Thus, it is continuous since $\max\{p_1, \dots, p_K\}$ is continuous in the probabilities. It does not satisfy I; consider the following counterexample: $[z_1] \sim [z_2]$, $[z_1] = 1/2[z_1] \oplus 1/2[z_1] \succ 1/2[z_1] \oplus 1/2[z_2]$.

Comparing the most likely prize: I will construct a counter-example. Let $\succ^*$ denote the basic preference relation over Z . Suppose that Z has two prizes and that $z_1 \succ^* z_2$. Furthermore, suppose that if there are any ties in the most likely prize, ties are broken in favor of z_1 . Then $[z_1] \succ [z_2]$ but $1/4[z_1] \oplus 3/4[z_2] \sim [z_2]$, so the preference relation does not satisfy independence. It is not continuous because $(1/2 - 1/n)[z_1] \oplus (1/2 + 1/n)[z_2] \sim [z_2]$ for all n but in the limit $(1/2)[z_1] \oplus (1/2)[z_2] \succ [z_2]$.

2. Suppose that such a utility function existed, that is, let $u : Z \rightarrow \mathbb{R}$ represents the preferences as described above. Since u is a utility that represents the preferences, then it must satisfy axioms the continuity and independence axioms and monotonicity for money axioms, plus any of the properties implied by each one of them. Towards a contradiction, we will suppose that there exists a utility function satisfying all the axioms, and from there we will derive a contradiction.

The preferences above imply that if u represents them:

$$\begin{aligned} u(600) &> u(400) > 0.90u(600) + 0.10u(0) \\ &> 0.20u(600) + 0.80u(0) > 0.25u(400) + 0.75u(0) > u(0). \end{aligned}$$

In particular notice that $[600]$ and $[0]$ are respectively the best and worst lotteries. Thus, $[400] \sim \lambda[600] + (1 - \lambda)[0]$ where $\lambda = \frac{u(400) - u(0)}{u(600) - u(0)}$, and $0.25[400] + 0.75[0] \sim \delta[600] + (1 - \delta)[0]$ where $\delta = 0.25\lambda$; in both cases we have used the continuity and independence axioms.

By monotonicity (monotonicity as implied by independence axiom, not monotonicity for money) we must have that $\lambda > 0.90$ and also that $0.20 > \delta = 0.25\lambda$. The last condition implies that $0.8 > \lambda$. Therefore, we have that $0.90 < \lambda < 0.8$ which is a contradiction.

This contradiction is happening because the independence axiom will always be violated. Consider the following: $[400] \succ 0.90[600] + 0.10[0]$. Suppose that we mix each side with the lottery $[0]$, with weights $(0.25, 0.75)$, if independence were satisfied, we would get that: $0.25[400] + 0.75[0] \succ 0.225[600] + 0.775[0]$ but this last lottery by monotonicity is strictly preferred to $0.20[600] + 0.80[0]$ which in turn is strictly preferred to $0.25[400] + 0.75[0]$ which is a contradiction. Thus the independence axiom always fails.

3. (a) $[1] \sim 0.5[0] \oplus 0.5[4]$ since $u([1]) = u(0.5[0] \oplus 0.5[4]) = 1$. However, for $\alpha = 1/2$:

$$\alpha[1] \oplus (1 - \alpha)[0.5[0] \oplus 0.5[2]] \succ \alpha[0.5[0] \oplus 0.5[4]] + (1 - \alpha)[0.5[0] \oplus 0.5[2]]$$

since the utility of the left lottery, $7/8$, is greater than the utility of the right lottery, $13/16$.

- (b) If you interpret L as a random variable over money:

$$\begin{aligned} u(L) &= \mathbf{E}[L] - (\mathbf{E}[L])^2 - \text{Var}[L] = E[L] - (\mathbf{E}[L])^2 - (\mathbf{E}[L^2] - (\mathbf{E}[L])^2) \\ &= \mathbf{E}[L] - \mathbf{E}[L^2] = \mathbf{E}[L - L^2] = \sum_{z \in Z} L(z)(z - z^2). \end{aligned}$$

Thus, if we define $v(z) := z - z^2$, we have that $u(L) = \sum_{z \in Z} L(z)v(z)$. Therefore, we have shown that u can be represented with an expected utility form using the function v . By proposition 3.13 in the slides, u is consistent with the vNM axioms.